SMALL DATA MACHINE LEARNING: FUNDAMENTAL CHALLENGES AND TARGETED SOLUTIONS

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ABSTRACT

Machine learning on small tabular datasets ($n < 1000$) presents unique mathematical bottlenecks, including severe overfitting, vulnerability to outliers, and the impossibility of standard hyperparameter tuning due to the lack of validation data. Instead of proposing a universal framework, this paper conceptualizes these fundamental challenges and presents two targeted solutions as case studies. For robust classification, we evaluate SmallGBM, which utilizes median leaf weights and stochastic split selection to mitigate outlier sensitivity without hyperparameter tuning. For nonlinear regression, we analyze SmallMLP, an adaptive neural network that derives its architecture directly from dataset parameters and employs weighted conformal prediction for calibrated uncertainty quantification. Through rigorous zero-tuning benchmarks across over 20 datasets, we establish the strict performance boundaries of these methods, demonstrating scaling curves that identify exactly where standard models like XGBoost and standard MLPs become optimal. By explicitly documenting both empirical successes and negative results, this work provides a grounded, mathematically rigorous foundation for future research in extreme low-sample learning regimes.

1 Introduction: The Small Data Paradigm

For the past two decades, the trajectory of machine learning has been overwhelmingly defined by the "big data" paradigm. Success stories in computer vision [**?**, **?**], natural language processing [**?**, **?**], and tabular prediction [**?**] are predicated on the availability of massive datasets where highly parameterized models can approximate arbitrarily complex functions. However, a vast array of critical scientific and industrial problems do not operate in this regime. In clinical medicine [**?**, **?**], materials science [**?**, **?**], and high-energy physics [**?**, **?**], data acquisition is inherently bottlenecked by physical or financial constraints, yielding datasets with fewer than 1000 samples. In these high-stakes domains, algorithms must adapt to strict data limits rather than relying on scale.

To address tabular data, a substantial body of literature has established gradient-boosted decision trees as the prevailing state-of-the-art. Implementations such as XGBoost [**?**], LightGBM [**?**], and CatBoost [**?**] consistently outperform alternative methods on medium-to-large tabular datasets [**?**, **?**]. These frameworks utilize second-order Taylor approximations for leaf weight estimation. While robust alternatives (e.g., Huber loss, mean absolute error) exist within these libraries, their default mean-based ERM estimators remain highly sensitive to heavy-tailed noise and outliers [**?**, **?**]. Crucially, selecting the appropriate robust loss function and configuring its thresholds—alongside coordinating depth and regularization—strictly depends on hyperparameter tuning via validation splits [**?**, **?**]. On scarce data, this validation process inevitably breaks down.

Parallel literature has extensively explored deep learning adaptations for tabular data. Recent advances include specialized attention mechanisms (e.g., TabNet [**?**], SAINT [**?**]), differentiable trees (NODE [**?**]), and heavily regularized multi-layer perceptrons [**?**, **?**]. While these models offer smooth function approximation, they require extensive neural architecture search, exacerbating the depletion of training data in low-sample regimes. Additionally, standard architectures lack native uncertainty quantification (UQ). While Bayesian Neural Networks [**?**], Deep Ensembles [**?**],

and Gaussian Processes [**?**] provide principled UQ, they frequently suffer from kernel degeneracy and Type-II maximum likelihood overfitting when dimensionality approaches the sample size ($d \approx n$).

Recently, in-context meta-learning models have been proposed to bypass validation tuning entirely. Most notably, Prior-Data Fitted Networks (TabPFN) [**?**, **?**] achieve strong zero-shot performance by performing approximate Bayesian inference via a Transformer pre-trained on synthetic causal models. However, TabPFN is structurally constrained by the quadratic $\mathcal{O}(n^2)$ computational complexity of its attention mechanism and imposes strict hard limits on feature dimensionality (typically $d \leq 100$) and class count. Concurrently, to address UQ without unverifiable parametric assumptions, conformal prediction [**?**, **?**] and its locally adaptive weighted variants [**?**, **?**] have emerged as rigorous methods for finite-sample coverage. In summary, while existing literature optimizes models for scale or synthetic priors, it lacks tuning-free, robust estimation for continuous regression and unconstrained classification in strict low-sample regimes.

This structural gap is rooted in the inherent limitations of standard model selection. When minimizing empirical risk on small data ($n < 1000$), extracting a validation set to constrain the hypothesis class dangerously depletes the scarce training signal [**?**, **?**]. The resulting selection bias and validation set overfitting dominate the optimization process [**?**], meaning the configuration that minimizes validation loss frequently exploits random sample fluctuations rather than identifying a true generalizable signal.

The aim of this paper is to formalize these fundamental bottlenecks of small tabular data and to present targeted, pre-tuned algorithmic baselines. In order to do this, we introduce two case studies: SmallGBM, which utilizes size-adaptive leaf aggregation and stochastic split selection for robust classification; and SmallMLP, an adaptive neural network that employs empirically validated capacity scaling rules based on dataset constraints (n, d, K) alongside weighted conformal prediction for calibrated uncertainty. Through rigorous zero-tuning benchmarks, we establish the strict performance boundaries of these methods, providing a grounded, empirically robust foundation for extreme low-sample learning.

2 Methodology and Experimental Setup

2.1 Meta-Training Protocol and the Avoidance of Data Leakage

To ensure a mathematically rigorous evaluation and strictly prevent data leakage, we separate the derivation of our algorithmic constants from the final benchmark. All heuristic hyperparameters for our methods—such as structural thresholds and capacity scaling limits—were derived empirically on a disjoint *Meta-Training Set* comprising a diverse collection of small tabular datasets sourced from the *scikit-learn* library and the UCI Machine Learning Repository, strictly disjoint from the evaluation benchmarks. The models were subsequently frozen. The evaluations reported in Section 3 are conducted on a completely independent *Hold-Out Benchmark Set* in a strict zero-tuning regime, ensuring that our methods cannot overfit to the test datasets at the meta-level.

2.2 SmallGBM: Robust Gradient Boosting

SmallGBM replaces the non-robust second-order Taylor approximation with a hybrid leaf estimator. For leaves with $|I| \leq 30$ (a threshold derived during the meta-training phase), we compute the sample median of the pseudo-residuals, leveraging its local 50% breakdown point to robustify leaf estimates against isolated noise spikes. For larger leaves, we transition to an inverse-distance weighted mean based on feature-space proximity to recover statistical efficiency on cleaner data. To prevent the additive tree ensemble from isolating individual heavy-tailed noise spikes, split selection is heavily stochastic: only five random thresholds per feature are evaluated. The core tree logic is implemented in a native C backend utilizing 256-bin histograms, yielding a $2\times$–$3\times$ computational speedup over native Python implementations.

2.3 SmallMLP: Adaptive Neural Networks and Tuning-Free Conformal UQ

SmallMLP eliminates manual architecture search by deriving the width of the first hidden layer deterministically from dataset parameters: $w_1 = \max(2K, \min(\lfloor \alpha\sqrt{nK\sqrt{d}} \rfloor, 4n))$, where the structural constant $\alpha = 4.0$ was fixed during meta-training. Subsequent layers decay geometrically by a factor of $\beta = 0.7$. Weight decay is scaled inversely with sample size ($\lambda = 1/n$), and dropout is clamped to $[0.1, 0.5]$.

For uncertainty quantification (UQ), we integrate locally weighted conformal prediction. To mitigate the validation depletion paradox previously discussed, we do not tune the Gaussian kernel bandwidth h_{cal} via grid search. Instead, we use the deterministic median heuristic: $h_{\text{cal}} = \text{median}(\|h(x_i) - h(x_j)\|)$, computed over the pairwise distances of the calibration set in the penultimate embedding space $h(x)$. While pairwise distances can concentrate in high

dimensions, the strict capacity bottlenecks of SmallMLP preserve sufficient metric structure. Although this framework still requires an initial calibration split, it resolves the tuning paradox: we entirely avoid iterative, high-dimensional hyperparameter validation splits (which induce selection bias and severe overfitting), utilizing a single, non-tuned split strictly to guarantee valid finite-sample marginal coverage.

2.4 Dataset Constraints and Mathematical Bottlenecks

To rigorously evaluate our methods, we selected hold-out benchmark datasets that natively exhibit the fundamental mathematical bottlenecks of small tabular data ($n < 1000$). Standard algorithms fail in these regimes due to the *Trilemma of Small Data Learning*: the inability to simultaneously achieve low variance, low bias, and calibrated uncertainty without depleting the training set for validation tuning.

We classify our evaluation datasets into three primary constraint domains:

- **Clinical Medicine ($d \approx n$ Regime):** High-dimensional, low-sample medical datasets (e.g., *Hepatitis*: $n = 155, d = 19$; *Breast Cancer*: $n = 569, d = 30$). This regime induces ill-conditioned covariance matrices $\hat{\Sigma}$ and destroys the validity of hyperparameter tuning, as standard validation splits critically deplete the scarce training signal.
- **Physical and Engineering Sciences (Heteroscedastic Noise):** Experimental physical datasets (e.g., *Yacht Hydrodynamics*: $n = 308, d = 6$; *Energy Efficiency*: $n = 768, d = 8$) often exhibit input-dependent noise variance $\sigma^2(x)$. Standard models that assume homoscedastic residuals miscalibrate their uncertainty, necessitating the locally adaptive conformal intervals of SmallMLP.
- **Physical Sensors and Remote Sensing (Heavy-Tailed Outliers):** Sensor data (e.g., *Ionosphere*: $n = 351, d = 34$; *Parkinson's*: $n = 195, d = 22$) frequently contains extreme measurement outliers. Because mean-based estimators have an asymptotic breakdown point of zero, these datasets strictly demand robust statistics (e.g., median-based leaf weights) to prevent catastrophic variance.

These three domains were selected because they map directly to the mathematical failure modes that SmallGBM and SmallMLP are designed to address. By evaluating on these specific constraints, we test whether our targeted solutions can overcome the Trilemma in practice.